

TITLE: Multi-Agent System for Automated Cardiac MRI
Analysis and Decision Support

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Abstract (Bahasa Melayu)

Penyakit kardiovaskular terus menjadi penyebab utama kematian di seluruh dunia, manakala pengimejan resonans magnetik jantung (Cardiac MRI) diiktiraf sebagai modaliti paling tepat untuk menilai morfologi jantung, fungsi jantung, dan pencirian miokardium secara tidak invasif. Walaupun modaliti ini menawarkan prestasi yang lebih baik berbanding kaedah lain, proses interpretasi Cardiac MRI masih bersifat berpecah-pecah, memerlukan prosedur yang panjang serta kepakaran tinggi daripada pakar kardiologi. Pelbagai penyelesaian berasaskan kecerdasan buatan (AI) telah dibangunkan bagi memperkemaskan aspek segmentasi, pengkuantitian, dan pelaporan imej jantung. Namun demikian, tiada penyelesaian sedia ada yang menyediakan interpretasi berdasarkan garis panduan klinikal yang dikeluarkan oleh European Society of Cardiology (ESC), American College of Cardiology (ACC), dan American Heart Association (AHA). Selain itu, kebanyakan sistem tersebut juga tidak menawarkan penjelasan klinikal yang menyokong keputusan diagnostik yang dihasilkan.

Sistem yang dicadangkan untuk interpretasi automatik imej Cardiac MRI menggunakan seni bina AI berasaskan multi-agen yang terdiri daripada empat agen yang berbeza tetapi saling berkoordinasi. Agen 1, iaitu Agen Segmentasi, melaksanakan segmentasi imej jantung menggunakan seni bina nnU-Net. Agen 2, dikenali sebagai Agen Analisis Kuantitatif, mengira parameter klinikal penting seperti Left Ventricular Ejection Fraction (LVEF), End-Diastolic Volume (EDV), End-Systolic Volume (ESV), dan jisim miokardium berdasarkan output daripada Agen Segmentasi. Agen 3, yang dinamakan Agen Penaakulan Klinikal, merupakan komponen paling inovatif dalam seni bina yang dicadangkan. Agen ini mendapatkan petikan berkaitan daripada korpus bukti klinikal yang terdiri daripada garis panduan ESC, ACC, dan AHA menggunakan teknik pencarian padat (dense retrieval) berasaskan BioBERT serta pengindeksan FAISS, sekali gus memastikan setiap kenyataan diagnostik disokong oleh bukti klinikal yang boleh dirujuk. Agen 4, iaitu Agen Kebolehhjelasan (Explainability Agent), menjana peta haba perhatian Grad-CAM dan mengintegrasikan sitasi garis panduan ke dalam laporan klinikal yang berstruktur.

Kerangka kerja ini dinilai menggunakan set data Cardiac MRI ACDC dan M&Ms yang merangkumi 525 subjek daripada pelbagai pusat klinikal, vendor, dan kategori patologi. Penilaian dilakukan menggunakan metrik Dice Similarity Coefficient (DSC), Mean Absolute Error (MAE) untuk parameter berasaskan isipadu, serta BERTScore bagi menilai kualiti laporan yang dihasilkan. Kajian ablasi turut dijalankan untuk mengenal pasti sumbangan individu komponen Explainable AI (XAI) dan lapisan orkestrasi agen terhadap prestasi keseluruhan sistem. Setakat pengetahuan penyelidik, sistem ini merupakan percubaan pertama yang menggabungkan orkestrasi multi-agen, penaakulan berasaskan garis panduan klinikal, dan kebolehhjelasan AI dalam satu kerangka interpretasi Cardiac MRI. Oleh itu, sistem yang dicadangkan berupaya menangani empat cabaran utama yang dikenal pasti melalui analisis sistematik terhadap lebih daripada 40 artikel penyelidikan yang diterbitkan antara tahun 2022 hingga 2026.

Kata Kunci: AI Beragen (Agentic AI), Kecerdasan Buatan Boleh Dijelaskan (Explainable AI), Cardiac MRI, Sistem Multi-Agen, Sokongan Keputusan Klinikal.

Abstract (English)

Cardiovascular diseases continue to pose the highest rate of mortality worldwide, with cardiac MRI being well acknowledged as the most accurate modality for evaluating cardiac morphology, function, and myocardial characterization noninvasively. Even though this modality outperforms other methods, cardiac MRI interpretation continues to be a fragmented process involving long procedures that require extensive cardiologist expertise. Artificial intelligence solutions have attempted to streamline and integrate different aspects of cardiac MRI, such as segmentation, quantification, and reporting. However, none of the existing solutions offer interpretations based on any of the clinical guidelines formulated by the European Society of Cardiology (ESC), American College of Cardiology (ACC), and American Heart Association (AHA); in addition, none provide clinical explanations.

The proposed system for automated interpretation of cardiac MRI images is a multi-agentic AI architecture, consisting of four distinct yet coordinated agents. Agent 1, which is the Segmentation Agent, executes cardiac image segmentation through nnU-Net architecture. Agent 2, known as Quantitative Analysis Agent, calculates the left ventricular ejection fraction, end-diastolic volume, end-systolic volume, and myocardial mass based on the outputs from the Segmentation Agent. Agent 3, referred to as the Clinical Reasoning Agent, is used to retrieve relevant passages from an evidence corpus consisting of ESC, ACC, and AHA guidelines by means of BioBERT-based dense retrieval and FAISS indexing technique, grounding all diagnostic statements in citable clinical evidence. Agent 4, the Explainability Agent, produces Grad-CAM attention heatmaps and integrates guideline citations into a structured clinical report.

The framework is evaluated based on ACDC and M&Ms; cardiac MRI data sets, which involve 525 subjects across various clinical centres, vendors, and pathology classes, using Dice Similarity Coefficient, mean absolute error for volume-related metrics, and BERTScore for report quality. Ablation studies isolate the individual contributions of the XAI component, and the agentic orchestration layer. To the best of the researcher's knowledge, this system is the first attempt at solving the problem of agentic orchestration along with guidelines-based inference and explainability in a single cardiovascular MRI interpretation framework, thus solving four crucial problems that were revealed during a systematic analysis of over 40 peer-reviewed articles from 2022-2026.

Keywords: Agentic AI, Explainable AI, Cardiac MRI, Multi-Agent Systems, Clinical Decision Support

Chapter 1: Introduction

1.1 Background:

According to the World Health Organization (2023), CVD is still the number one killer of people across the globe, causing about 17.9 million deaths each year, thus accounting for about 32% of all deaths recorded. Cardiac disorders continue to impose a significant burden on the healthcare sector due to ageing population demographics and increased incidences of diseases such as hypertension, diabetes, and obesity.

Out of the various techniques for cardiac investigation, cardiac magnetic resonance imaging (CMR) has been considered the gold standard method for non-invasive evaluation of cardiac morphology, cardiac function, and myocardial tissue characterization (Kilner et al., 2010). CMR is known to have higher reproducibility in estimating left ventricular ejection fraction (LVEF), end-diastolic volume (EDV), and end-systolic volume (ESV) than echocardiography, and is the most preferred imaging tool for identifying myocardial fibrosis, cardiomyopathy, and heart structure diseases. The clinical practice guidelines formulated by the European Society of Cardiology (ESC), American College of Cardiology (ACC), and American Heart Association (AHA) clearly indicate that CMR plays an important role in the diagnosis and risk assessment of patients suffering from dilated cardiomyopathy (DCM), hypertrophic cardiomyopathy (HCM), and ischemic heart diseases.

Although clinically superior, the clinical adoption of CMR is limited in terms of workflow constraints. A complete CMR scan involves the acquisition of many images in different sequences such as cine, late gadolinium enhancement (LGE), and parametric mapping scans. Each set of images needs to be interpreted. This process is time-consuming and prone to bias. Due to the global deficit of specialists such as cardiologists and radiologists as well as increased numbers of imaging, there is a high demand for intelligent algorithms that can assist in CMR interpretation at scale (Rajpurkar et al., 2022).

Artificial intelligence (AI), especially deep learning techniques, have shown great potential in automating tasks associated with CMR imaging independently. Convolutional neural networks have shown almost expert-level accuracy in heart segmentation and estimating the ejection fraction. More recently, AI foundation models pre-trained on CMR datasets like CineMA (Fu et al., 2025) for cine CMR, as well as ScarNet (Tavakoli et al., 2025) for LGE-based scar quantification, have enhanced single task automation. Despite their success, these models work individually, targeting individual tasks in the CMR analysis pipeline rather than incorporating them into clinical workflows (Litjens et al., 2017).

An entirely new paradigm has emerged with the advent of agentic artificial intelligence, whose hallmark lies in autonomous planning, reasoning, and action in executing multi-stage tasks, with the use of specialized instruments or sub-agents (Xu et al., 2025). When applied alongside Retrieval-Augmented Generation, whose purpose is to anchor the outputs of AI in external knowledge sources such as clinical guidelines, the resulting combination has the promise of bringing about fully end-to-end diagnostic reasoning processes that are precise, traceable, and clinically understandable. Recent review articles on agentic AI in radiology and medicine highlight the paradigm as one that holds great promise but remains underexplored, especially when it comes to cardiac MRI procedures (Wind et al., 2025; Khosravi et al., 2026).

1.2 Problem Statement:

The existing AI implementations for cardiac MRI are currently suffering from several limitations related to fragmentation, meaning that their functionality includes just one modality, task, or type of disease, and often represents opaque algorithms that value technical performance rather than clinical utility. Such fragmentation causes major challenges for clinicians who are responsible for managing separate processes of manual segmentation, quantitative assessment, analysis between sequences, and report creation, all of which require substantial amounts of time and expertise and involve inter-rater variability (Leiner et al., 2019).

This research is being conducted due to the following three issues:

First, fragmented task automation. Current CMR AI algorithms are built for single tasks. CineMA (Fu et al., 2025) is a foundation model capable of segmentation tasks but not extending further into clinical reasoning. Report generation tools like MARL-Rad (Baba et al., 2025) create structured reports, though these are not based on imaging findings. Multimodal dialogue management algorithms like AURA (Fathi et al., 2025) and MedChat (Liu et al., 2025) work with multimodal dialogues but lack a pipeline for cardiac-specific imaging. There are no currently available algorithms integrating these components into an end-to-end clinical decision-making process. Consequently, clinicians still have to combine disparate AI-produced outputs manually.

Second, the absence of guideline adherence. State-of-the-art agents specialized in cardiovascular diagnostics, BAAI Cardiac Agent (Qu et al., 2026) and CardAIC-Agents (Zhang et al., 2025), deliver diagnostic outputs without reference to clinical guidelines issued by the ESC, ACC/AHA, or the Heart Failure Society of America (HFSA). In high-pressure medical settings, outputs not backed up by any criteria defined in guidelines are of little use to practicing cardiologists and radiologists. The only known example of a guideline-based algorithm, which connects clinical narratives with ACR Appropriateness Guidelines (Pambudi et al., 2025), works only with general radiology text and has not been extended to cardiac imaging.

Third, insufficient explainability. Multiple systematic reviews on the explainability of cardiac AI from 2025 uniformly report that most AI tools used in the field of cardiac imaging are not sufficiently explainable (Haupt et al., 2025; Salih et al., 2025). Specifically, two systematic reviews report that the number of explainable CMR models is very limited and that the methods of explainable AI lack validation on criteria that would ensure their usefulness in practice. The lack of explainability poses substantial risks in terms of patient safety since any incorrect classification of a cardiomyopathy or missed myocardial scar could alter the treatment decision with severe consequences for patients (Rudin, 2019).

This research addresses these three compounded problems by proposing the design, development, and evaluation of a retrieval-augmented agentic AI framework that delivers

explainable, guideline-grounded cardiac MRI interpretation within a unified multi-agent pipeline, advancing both the science of agentic medical AI and its practical applicability in clinical cardiology.

1.3 Research Gaps:

The gaps uncovered through a systematic review of more than 40 peer-reviewed articles from 2022 to 2026, including studies on agentic AI surveys, multimodal medical AI, cardiac imaging with deep learning, clinical systems with explainable AI for medical imaging, are as follows: All four gaps are significant and interrelated; no single system addresses all of them at once.

Gap Type	Evidence from Literature	Research Opportunity
Knowledge Gap	Surveys of agentic AI in healthcare (Xu et al., 2026) confirm a notable scarcity of dedicated agentic systems for cardiac MRI. Healthcare agents are largely domain-generic and lack imaging-specific orchestration workflows.	Design a dedicated cardiac MRI agentic framework integrating automated segmentation, quantitative analysis, and guideline-grounded clinical reasoning.
Methodological Gap	Multimodal agents, AURA (Fathi et al., 2025), MedChat (Liu et al., 2025), and MMedAgent-RL (Xia et al., 2025), operate on isolated tasks without unified end-to-end pipelines required for complex clinical diagnosis.	Integrate segmentation, clinical reasoning into a single orchestrated multi-agent pipeline.
Experimental Gap	Cardiac-specific agents, including BAAI Cardiac Agent (Qu et al., 2026) and CardAIc-Agents (Zhang et al., 2025), lack validation on multi-centre or heterogeneous datasets. Many systems are conceptual or insufficiently validated on real-world data.	Validate the framework on ACDC and M&Ms; datasets using standardised metrics for diagnostic accuracy and workflow alignment.
Clinical Gap	Systematic XAI reviews in cardiovascular imaging (Haupt et al., 2025; Salih et al., 2025) confirm most deployed systems produce opaque predictions misaligned with ACC/AHA and ESC CMR clinical guidelines.	Embed guideline-grounded outputs and attention-based explanations to produce transparent, traceable clinical decision support.

Table 1.1. Summary of identified research gaps and corresponding opportunities.

Together, the four gaps constitute a strong justification for the novel contribution to research that will be made from both the level of knowledge creation as well as methodology, experimentation, and clinical practice.

1.4 Research Questions:

The following research questions have been identified to guide the development of the proposed framework:

RQ1. How effectively can a multi-agent pipeline integrate automated segmentation, quantitative cardiac analysis, and guideline-grounded clinical reasoning within a unified CMR workflow, as evaluated by task-specific and end-to-end performance metrics?

RQ2. How does the proposed framework improve diagnostic accuracy and provide clinically meaningful explainable reasoning compared with established single-task AI baselines using standardized metrics (Dice, MAE, and BERTScore)?

RQ3. How well does the proposed framework generalise across heterogeneous cardiac MRI datasets from multiple acquisition protocols and clinical centres, as assessed by segmentation accuracy and diagnostic consistency?

1.5 Research Objectives:

The following objectives operationalise the research questions into concrete, measurable contributions:

O1. To design and implement a retrieval-augmented agentic AI framework for automated cardiac MRI interpretation, comprising specialised agents for segmentation, quantitative analysis, guideline-grounded reasoning, and explainable output generation.

O2. To integrate explainable clinical reasoning into the proposed framework and benchmark its clinical accuracy and explanation quality against single-task baselines using Dice Similarity Coefficient (DSC), Mean Absolute Error (MAE), and BERTScore metrics.

O3. To validate the robustness and generalisability of the proposed framework across heterogeneous publicly available cardiac MRI datasets, including the ACDC dataset and the Multi-Centre, Multi-Vendor, and Multi-Disease Cardiac Segmentation (M&Ms) challenge dataset.

1.6 Research Scope:

This research work has its scope defined to make sure that the study remains feasible for completion in the Master of Computer Science by Research program offered by Universiti Malaya. Table 1.2 illustrates the scope of inclusion and exclusion in the current research.

In Scope	Out of Scope
Cine cardiac MRI (primary modality)	LGE and T1 mapping modalities (future work)
Left ventricular segmentation; computation of LVEF, EDV, ESV	Right ventricular strain and myocardial deformation analysis
XAI module grounded in ESC CMR 2022 and ACC/AHA guidelines	Real-time clinical deployment or hospital PACS integration
Publicly available datasets: ACDC and M&Ms;	Proprietary or single-institution hospital datasets
Explainability via attention-based visualisation and guideline citation	Pixel-level SHAP or LIME explanations on raw MRI data
Automated structured clinical report generation	Conversational clinical dialogue or patient-facing interfaces

Table 1.2. Research scope boundaries.

This study mainly emphasizes cine cardiac MRI as the primary imaging modality due to its high functional evaluation information and extensive availability in public datasets. Other modalities like LGE and parametric mapping will add another layer of complexity to preprocessing outside the capabilities of the scope of this research and will be considered in the future. The RAG module will extract information from a selected collection consisting of ESC 2022 CMR guidelines and cardiac imaging positions by ACC/AHA. The experiments will be performed using open-source and ethically available datasets ACDC (Bernard et al., 2018) and M&Ms (Campello et al., 2021) that provide ground truth expert segmentations for various cardiac diseases.

1.7 Research Significance:

This research makes significant contributions at three levels.

Scientific significance. To the best of the researcher's knowledge, no prior work has addressed agentic orchestration, retrieval-augmented guideline grounding, and explainability simultaneously for cardiac MRI reading in a single integrated system before. Through addressing all four gaps found during the systematic literature analysis directly, the paper brings further advancements to the field of medical agentic AI (Xu et al., 2026), clinical reasoning powered by RAG (Pambudi et al., 2025), and explainability in cardiac imaging (Haupt et al., 2025; Salih et al., 2025). The proposed framework brings a unique architectural solution and evaluation approach to benchmark results for further research in cardiac AI and agentic medical systems more broadly.

Clinical significance. By producing guideline-cited, attention-visualised, and quantitatively validated outputs, the proposed system directly supports the decision-making workflow of

radiologists and cardiologists interpreting CMR studies. Aligning the outputs generated by AI applications with recommendations included in the clinical guidelines for CMR developed by the ACC/AHA and the ESC addresses the problem of disconnect between the predictions made by AI algorithms and evidence-based recommendations that are mandatory for medical practitioners to use. The multi-agent architecture mirrors the multi-disciplinary nature of real cardiac imaging workflows, making the system conceptually familiar and practically integrable into clinical environments.

Societal significance. Access to specialist cardiac imaging expertise is unequally distributed globally, with significant shortfalls in low- and middle-income countries and non-urban healthcare settings (Roth et al., 2020). Scalable, AI-assisted CMR interpretation that is both accurate and explainable has the potential to reduce diagnostic bottlenecks, support non-specialist clinicians in resource-constrained environments, and contribute to more equitable access to high-quality cardiac diagnostic services. By demonstrating that AI-assisted cardiac reasoning can be grounded in international clinical guidelines, this research also contributes to the harmonisation of cardiac imaging practice across diverse health systems.

To conclude, the contribution made by this research is timely and relevant, scientifically innovative, clinically useful, and well-scoped.

Chapter 2: Literature Review

2.1 Introduction

This chapter presents a critical literature review of the scientific basis for the suggested retrieval-agented AI paradigm in cardiac MRI interpretation. The review is structured thematically into seven related topics, including: (1) cardiac MRI applications in medicine and AI automation; (2) deep learning approaches for cardiac MRI segmentation and quantitative analysis; (3) agentic AI models and applications in healthcare and radiology; (4) multimodal medical AI agents; (5) retrieval-augmented generation techniques in clinical decision support; (6) explainability in cardiac and medical imaging AI solutions; and (7) specific AI-based paradigms for cardiac MRI analysis. The sections follow a similar analytical framework, which includes an overview of existing knowledge, discussion of the problem area gap, and clarification of how the proposed research will contribute to the problem solution. This chapter closes with a synthesized table that compares various aspects of all four research gaps considered in this paper and their critical summaries.

2.2 Cardiac MRI in Clinical Practice and AI Automation

2.2.1 Clinical Role and Diagnostic Importance of CMR

Cardiac magnetic resonance imaging (CMR) is currently considered the gold standard of the non-invasive investigation of the heart's morphology, physiology, and tissue characterization. The superior performance of CMR compared to echocardiography in estimating (LVEF), end-diastolic volume (EDV), and end-systolic volume (ESV) was well-documented in numerous studies (Leiner et al., 2019). CMR enables multi-sequence examination including the functional cine imaging, late gadolinium enhancement (LGE) for myocardial scarring, and T1/T2 parametric mapping for diffuse myocardial disease. It is recognized by clinical practice guidelines developed by the ESC, ACC, and AHA as the method of choice for diagnosing dilated cardiomyopathy (DCM), hypertrophic cardiomyopathy (HCM), myocarditis, and cardiac sarcoidosis.

However, due to its complexity, interpreting CMR data is still associated with high workloads that can be addressed using artificial intelligence. A comprehensive CMR study includes hundreds of images that need to be assessed, while manual segmentation and volume quantification requires significant time and is subject to up to 10% inter-operator variability for LVEF assessments (Leiner et al., 2019). In addition, there is a global shortage of experienced cardiologists in interpreting CMR studies, creating an urgent need for intelligent automated support.

2.2.2 Deep Learning for CMR Segmentation and Quantification

Deep learning applications in CMR segmentation have evolved significantly since the advent of the ACDC challenge (Bernard et al., 2018), which assessed the accuracy of convolutional neural networks for multi-class cardiac segmentation. More recently, it was shown that the use of U-Nets allowed obtaining Dice Similarity Coefficients above

0.90 in the segmentation of left ventricular myocardium, achieving the same level as the inter-observer variability. In the more recent M&Ms; challenge (Campello et al., 2021), researchers demonstrated that even though the accuracy obtained was good, generalization among different acquisition settings is yet to be achieved.

The use of foundation models has further pushed the limits of CMR automation in deep learning. For instance, CineMA (Fu et al., 2025) presented a pre-trained foundation model for cine CMR that achieved state-of-the-art results for segmentation and left ventricular ejection fraction estimation, without needing fine-tuning for any specific task across different datasets. In another case, ScarNet (Tavakoli et al., 2025) presented a pre-trained model that was designed for the automatic quantification of myocardial scars in LGE-CMR scans. A systematic review assessing the use of artificial intelligence in cardiac MRI for coronary artery disease (Jiménez-Jara et al., 2025) found that although models performed well in localised tasks but lack integration into diagnostic workflows. Similarly, a review of advanced CMR integrating AI (Malagi et al., 2025) noted that while reconstruction and segmentation have matured, clinical adoption remains limited by the absence of explainability and workflow integration.

A critical limitation with all the aforementioned techniques of segmentation and quantification is their isolation in terms of task, and lack of integration with any other part of the diagnosis process such as clinical reasoning, guideline integration, and explanation generation. Although CineMA and ScarNet offer sophisticated solutions for segmentation masks and scalar measurements respectively, there is no integration of these tools with clinical guidelines and diagnostic explanation generation.

2.3 Agentic AI in Healthcare and Radiology

2.3.1 Foundations of Agentic AI

The concept of agentic AI involves the ability to plan, reason, and execute actions in multiple steps using the coordination of specialized tools or sub-agents (Xu et al., 2025). Contrary to traditional forms of AI, where only one output is created for one input, agentic AI entails task decomposition into smaller parts, selecting appropriate tools for performing each part of the action, and further refinement of each output based on intermediate outcomes. LLMs' capabilities as reasoning engines have played an essential role in developing such types of AI.

Several comprehensive surveys have analyzed the agentic AI in healthcare. According to Xu et al. (2026), more than 200 papers have been reviewed, revealing perception, reasoning, action, and memory as the four key cognitive components of healthcare agents. In the context of the use of AI agents in clinical practice (Gorenshtein et al., 2025), despite the promise of automated clinical workflows shown by the agentic systems, none of the analyzed systems have been validated clinically yet. Next-generation healthcare agentic AI was the focus of a forward-looking paper written

by Rajpurkar et al. (2025); they stressed the importance of agent specialization for particular domains, which is the goal of this study.

2.3.2 Agentic AI in Radiology

Agentic AI has been increasingly considered in radiology applications. In their recent study on agentic LLMs for radiology, Wind et al. (2025) highlighted report generation, clinical quality assurance, and workflow automation as the leading use cases, alongside the problems of hallucination and deployment. Khosravi et al. (2026) examined the developments in the field of radiology AI, from the emergence of static LLMs to autonomous agents, which can undertake multi-level clinical thinking, indicating that although there is a shift towards full-fledged agentic radiology AI, few studies have demonstrated clinical deployment. A review on the designs of agentic systems in radiology (Chen et al., 2025) identified the lack of benchmark measures for agentic system assessment as one of the major hurdles.

Significantly, no reviewed radiological agentic systems were developed to address specific issues related to cardiac MRI. The imaging processes included mostly those related to chest x-ray, general radiology report generation, and clinical QA based on textual data. The cardiac MRI analysis pipeline involves multiple stages, including various sequences and disease-specific guideline criteria for quantitative assessment. Therefore, this represents the knowledge gap described in Chapter 1.

2.4 Multimodal Medical AI Agents

2.4.1 General Multimodal Agent Frameworks

Multimodal AI agents that can interpret and analyze both image and text information represent a hot topic in contemporary scientific studies. For instance, the AURA (Fathi et al., 2025) was an interesting proposal for multi-modal medical agents capable of understanding, reasoning and annotating heterogeneous medical data sets. The AURA performed well in multimodal medical benchmarks but clearly identified its limitations in relation to cardiac image interpretation and lack of 4 diagnostic pipelines focused on particular diseases. MedChat (Liu et al., 2025) developed a novel multi-agents approach in multimodal disease diagnosis based on LLMs enabling clinical dialogues between agents for imaging and textual data; still, MedChat's capabilities were limited to visual question answering only.

Furthermore, the MMedAgent-RL (Xia et al., 2025) provided interesting insights into multimodal medical agents by suggesting multimodal medical reasoning through reinforcement learning in order to optimize the collaborative process of agents; however, its application is limited because of high computational requirements and lack of cardiac specific results. Med-Agent (Dąbrowicki et al., 2025) suggested a multi-modal hybrid approach in diagnosing oncological diseases with imaging and genomic data as input data but was not designed for use in cardiology. Lastly, Choi et al. (2025) discussed the potential of multimodal AI and agentic systems, identifying the model-context protocol as a promising integration standard but presenting no experimental validation.

2.4.2 Radiology Report Generation Agents

Report generation is considered one of the most explored applications of multimodal agents in medical imaging. Specifically, MARL-Rad (Baba et al., 2025) presented a multi-modal multi-agent reinforcement learning system that models radiologist's work to generate structured reports with improved BLEU and ROUGE scores. However, MARL-Rad is applied to general radiology report generation and does not employ guideline grounding, as well as explanation generation. Furthermore, RadCouncil (Zeng et al., 2025) leveraged a multi-agent language model approach for radiology impression generation improvement, with text-only input rather than image analysis being provided. Moreover, the Medical AI Consensus framework (Elboardy et al., 2025), which involved multi-agent debate in report generation and evaluation, achieved better factual accuracy but was non-specific to cardiology and had no real-world evaluation. Lastly, AgentsEval (Fu et al., 2025) offered an evaluation tool for medically faithful reports by means of multi-agent reasoning, thus contributing to evaluation but not offering an end-to-end diagnosis pipeline.

Notably, all of the aforementioned approaches have two key weaknesses – first, they do not implement any guideline grounding, meaning that their outputs are assessed in terms of lexical metrics only (e.g., BLEU, ROUGE, BERTScore), and second, they fail to establish a unified imaging-to-report pipeline.

2.5 Multi-Agent Systems for Medical Image Analysis

There is an increasing trend of the usage of Multi-agent systems and Multi-Agent Reinforcement Learning (MARL) techniques on various medical image processing tasks. A recent study that reviewed the use of multi-agent methods in medical image segmentation (Bennai et al., 2025) applications based on agent communication type, reward function type, and task type concluded that although MARL methods offer comparable performance for segmentation tasks, but there exists a need for standard benchmarking and clinical validation.

TissueLab (Li et al., 2025) designed a novel agentic AI model for medical imaging analysis through co-evolutionary processes, generating interactive imaging pipelines; however, testing is only done on pathology and histological tasks with no application in cardiology. In another application, M³Builder (Feng et al., 2025) developed an efficient method of automating machine learning workflows on medical imaging using a multi-agent model; nevertheless, there was no clinical-grade evaluation in their tests. Finally, a recent project on the development of an AI driven multi-agent system for radiologist workflow (Kristijan et al., 2025) has shown potential for workflow automation; however, they did not consider explainability.

In MARL-based segmentation approaches, several frameworks have achieved significant technical success. MARL-MambaContour (Zhang et al., 2025) used deep reinforcement learning in the field of active contour segmentation, whereas boundary-aware supervoxel segmentation by Ma et al., 2023 achieved enhanced accuracy in 3D medical image segmentation. The problem of heterogeneous data in distributed medical imaging was solved using fedMRL (Sahoo et al., 2025) based on federated multi-agent deep reinforcement learning. It is clear that none of these systems can extend to cardiac MRI and clinical reasoning as the proposed pipeline does.

2.6 Explainable AI in Cardiac and Medical Imaging

2.6.1 The Need for Explainability in Cardiac AI

Explanation in AI is a capability of the system in providing understandable justifications behind its predictions (Rudin, 2019). For AI applications in medical imaging, explainability is not an additional component but rather a necessity; doctors have to comprehend AI findings, review them critically, and use them for decision-making purposes. As per regulatory requirements of the EU AI Act and FDA guidance on AI/ML-based software as a medical device, explainability is becoming a requirement for clinical AI tools.

Two systematic reviews of explainable AI in cardiac and cardiovascular imaging provide an overview of the present situation. Haupt et al. (2025) examined explainable AI in cardiovascular radiological imaging and identified attention-based approaches, Grad-CAM, and SHAP as the most frequently used explainable models, while pointing out the scarcity of clinically implemented explainable AI tools and validation of explanations according to certain clinical utility criteria. Salih et al. (2025), in their study devoted exclusively to cardiac imaging explanation, reported similar findings: visual saliency maps were generated by most of the explainable AI models under investigation without using language consistent with clinical guidelines and diagnostic criteria.

2.6.2 Applied XAI in Cardiac AI Systems

In the applied systems literature, Kolk et al. (2024) have proposed an Explainable AI system that uses cardiac MRI imaging together with clinical information to find out which patients with non-ischaeamic cardiomyopathy are prone to having lethal ventricular arrhythmias. Even though it successfully showed that the combination of imaging and clinical factors can be helpful for the process of patient risk stratification, it has only focused on one class of diseases, and its explanations have been not grounded in the information provided in clinical guidelines. Wang et al. (2024) have developed a cardiovascular AI system for screening and diagnosing various types of diseases but also mentioned their lack of explainability explicitly. Concept-RuleNet (Sinha et al., 2026) combines neuro-symbolic reasoning with vision-language models to provide concept-level explanations and is still at a testing stage.

It has become apparent during the analysis of XAI literature that none of the available approaches provides explanations that can be grounded in passages from clinical guidelines. The proposed framework solves this problem by developing two modes of explanations, visual and textual, where the first will refer to attention maps of image segmentation outputs, and the second one to textual explanations given by a RAG.

2.7 Cardiac-Specific AI Agent Systems

There are two particular systems from the reviewed literature that approach the scope of the presented framework the closest and require comprehensive critical analysis.

BAAI Cardiac Agent (Qu et al., 2026) is one of the most innovative, technologically sophisticated cardiac agentic systems presented in the literature to date. It presents a novel end-to-end multi-modal framework for cardiac MRI reasoning and diagnosis, combining image analysis techniques with clinical text mining. BAAI Cardiac Agent has shown excellent performance on all evaluated benchmarks, thus significantly outperforming other

state-of-the-art solutions designed for single task processing. Nevertheless, the key limitations of BAAI Cardiac Agent are: (1) the evaluation is done only using the small dataset and no cross-validation on multi-centre/multi-vendor generalization set; (2) there is no information about knowledge retrieval and output alignment with ACC/AHA, ESC clinical guidelines; and (3) no explainability methods are employed, hence making it hard for physicians to interpret the results obtained by this system.

The CardAIC-Agents framework by Zhang et al., 2025 introduced a multimodal system with hierarchical adaptation for cardiac care support and used specialized agents for various cardiac data modalities. Although hierarchical architecture is an architectural feature shared by both models, the main issues in CardAIC-Agents are related to a lack of implementation studies, absence of guideline-based reasoning and lack of explainability. Evaluation of CardAIC-Agents is limited to one institution only.

Thus, while the difference lies in agentic architecture, the crucial differences between the two frameworks lie in three aspects, namely (1) guideline-based reasoning that references ESC/ACC guidelines in output; (2) explainability that establishes links between AI output and existing clinical knowledge; and (3) evaluation using multiple public datasets across multiple medical institutions.

2.8 Synthesis and Critical Summary

Comparison of relevant systems reviewed relative to the proposed model according to certain critical attributes is provided in Table 2.1. The last column indicates the uniqueness of the proposed model in addressing the four research gaps discussed in Chapter 1.

System	Year	Modality	Method	Limitation	Gap Addressed
CineMA	2025	Cine CMR	Foundation Model	No explainability; no clinical reasoning	Segmentation only
ScarNet	2025	LGE MRI	Foundation Model	Scar detection only; no guideline output	Single task
BAAI Cardiac Agent	2026	Cardiac MRI	Multimodal Agent	Limited multi-centre validation	Experimental Gap
CardAIC-Agents	2025	Cardiac Multimodal	Hierarchical Agents	No guideline grounding; limited deployment	Methodological Gap
AURA	2025	Multimodal Medical	Multimodal Agent	No cardiac-specific pipeline	Domain Gap
MedChat	2025	Multimodal Medical	Multi-Agent LLM	Limited imaging reasoning	Methodological Gap

MMedAgent-RL	2025	Multimodal Medical	RL Multi-Agent	High computational cost; no CMR focus	Methodological Gap
MARL-Rad	2025	Radiology Reports	MARL	No imaging grounding; no guidelines	Clinical Gap
ACR RAG System	2025	Radiology Text	RAG Multi-Agent	Text only; no cardiac imaging	Knowledge Gap
Proposed Framework	—	Cine CMR	Multi-Agentic AI + XAI	—	All four gaps

Table 2.1. Comparative analysis of relevant systems. Green row indicates the proposed framework.

Table 2.2 summarises the literature review findings by thematic category, mapping each theme to its collective limitation and the corresponding gap type.

Theme	Key Systems Reviewed	Collective Limitation	Gap Type
Cardiac MRI AI	CineMA, ScarNet, AI for CAD review, Advanced CMR review	Single-task, no unified pipeline, no guideline grounding	Knowledge + Methodological
Cardiac AI Agents	BAAI Cardiac Agent, CardAIC-Agents	Limited multi-centre validation, no explainability	Experimental + Clinical
Multimodal Medical Agents	AURA, MedChat, MMedAgent-RL, Med-Agent	Isolated tasks, no cardiac imaging pipeline	Methodological
Medical Imaging Agents	TissueLab, M³Builder, XrayClaw, MARL segmentation systems	No CMR specificity, no clinical reasoning	Domain + Experimental
RAG in Clinical AI	ACR RAG System, Agentic RAG for Radiology QA	Text-only, no cardiac imaging extension	Knowledge + Clinical
XAI in Cardiac Imaging	Haupt et al. review, Salih et al. review, Multimodal XAI for arrhythmia	No deployed explainable CMR systems; no guideline alignment	Clinical
Healthcare Agentic AI	Xu et al., Wind et al., Khosravi et al.	Generic frameworks; no cardiac MRI implementation	Knowledge

Table 2.2. Thematic summary of literature review findings and gap mapping.

The review reveals a consistent and multi-dimensional gap at the intersection of cardiac MRI AI, agentic systems, RAGs for clinical reasoning, and explainability. Among all the thematic areas surveyed, there currently exists no single framework that simultaneously: (1) develops an agentic pipeline for cardiac MRI, (2) integrates segmentation, quantitative analysis, and clinical reasoning, (3) retrieves from and aligns with clinical guidelines, and (4) generates explainable outputs using clinically validated criteria. The proposed framework aims to bridge this exact intersection.

It is significant to mention that the literature provides solid building blocks from which the proposed framework can benefit from: CineMA and ACDC models provide a proven basis for segmentation; LangChain and LlamaIndex provide a proven basis for RAGs; and the recent ACR RAG framework for medical AI (Pambudi et al., 2025) proves the feasibility of guideline grounding in the medical field. Thus, the present contribution is not about recreating the wheel, but about being the first to unite these building blocks into an integrated cardiac agentic pipeline, which is a truly novel contribution to the literature surveyed.

Chapter 3: Methodology

3.1 Research Design and Paradigm

The current research utilizes the paradigm of Design Science Research (DSR), which represents the most appropriate epistemological standpoint for studies focused on developing and testing new artefacts, in this case, an artificial intelligence system designed to be used as a solution for a particular class of problems in a specified field of study (Hevner et al., 2004). Design science research involves such phases as designing/implementing the artefact (build cycle) and evaluating its effectiveness relative to other solutions (evaluate cycle).

In the current research, the quantitative experimental methodology is utilized, whereby the performance of the system is assessed based on segmentation accuracy, quantitative measurement error and reporting quality through the use of publicly available MRI images from the heart annotated by experts. All of the aforementioned criteria are measured objectively and in a repeatable manner (Rajpurkar et al., 2022).

The research design involves the following four stages of process: (1) Foundation Stage, consolidation of literature, preparation of the dataset, and training of the backbone for the segmentation task; (2) Construction Stage, creation of an ensemble of agents, and integration of the XAI component; (3) Evaluation Stage, benchmarking against the baselines using ACDC and M&Ms datasets; and (4) Validation Stage, generalisation to new dataset, execution of ablation studies.

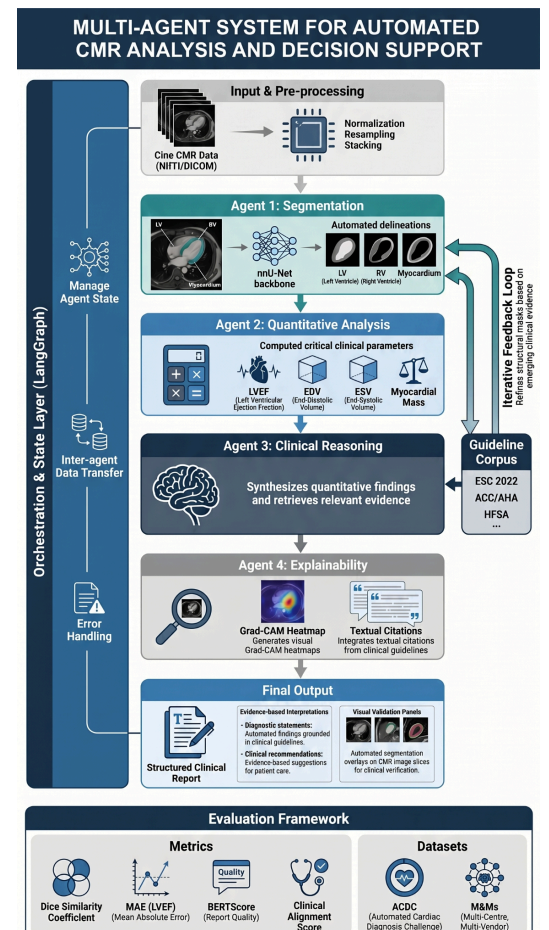
3.2 Conceptual Framework and System Architecture

The suggested architecture is a four-agent orchestration pipeline based on the modular specialization principle of unified reasoning: each agent is specialized in a particular domain for completing one well-defined task, and the orchestration layer ensures coherent passage of the output from one agent to another, leading to the production of a unified, guideline-driven, and interpretable clinical report. The system architecture is shown in Figure 3.1 below.

The framework consists of four agents and an orchestration module, which utilize cine cardiac MRI as the main imaging modality input. In addition, this system was designed to be modular, individual agents could be upgraded (for example, swapping the segmentation backbone with another one) without re-designing the entire framework. The orchestrator takes care of agent states, inter-agent data flows, and errors throughout pipeline operation.

Four specialized agents are employed:

Agent 1: Segmentation, automated delineation of cardiac structures;



Agent 2: Quantitative Analysis, calculation of clinical parameters of the heart based on results provided by segmentation;

Agent 3: Clinical Reasoning, the main novel agent that selects relevant passages from clinical guidelines and creates the rationale for diagnoses;

Agent 4: Explainability, creation of visualization attention maps and citing guidelines in the output.

All agents are detailed in Sections 3.4-3.7.

3.3 Agent 1: Segmentation Agent

3.3.1 Design Rationale

The Segmentation Agent is tasked with the automatic segmentation of cardiac regions in cine CMR image sequences, generating binary masks for LV cavity, RV cavity, and LV myocardium. The Segmentation Agent is not itself a novelty claim by itself but instead offers the necessary imaging base that supports the novel reasoning processes (agents 3 and 4). The use of a proven, highly efficient segmentation back end guarantees that any errors will be reduced and allows the evaluation of the impact of XAI components to be cleanly made.

3.3.2 Segmentation Backbone

The segmentation backbone follows nnU-Net (Isensee et al., 2021), which is an auto-configurable neural network that dynamically reconfigures itself to match the requirements of any particular medical imaging dataset. nnU-Net was chosen among the several available frameworks (such as TransUNet, SwinUNETR) for three reasons: (1) it performs state-of-the-art or near-state-of-the-art results on the ACDC leaderboard without any adaptation to the specificities of the dataset; (2) it allows for building a reproducible workflow to compare different models against the baseline models reported in the literature; and (3) it is a native implementation within the PyTorch library. The cineMA (Fu et al., 2025) approach will be tested as a zero-shot approach to the segmentation task in the ablation study.

3.3.3 Input Preprocessing

The raw cine CMR data is available in NIfTI format from both ACDC and M&Ms databases. The pre-processing includes: (i) resampling of voxels into a common in-plane resolution of 1.5×1.5 mm and 8 mm slice thickness; (ii) z-score intensity normalisation for each volume; (iii) random patch extraction of 192×192 patches during training and central cropping during inference; and (iv) selection of ED and ES frames based on time gradient analysis since both frames are necessary to calculate the volume by Agent 2.

3.4 Agent 2: Quantitative Analysis Agent

The Quantitative Analysis Agent takes in binary segmentation masks produced by Agent 1 and produces the standard clinical cardiac parameters. Left ventricular end-diastolic volume (EDV) and end-systolic volume (ESV) are calculated by adding up the volume of all voxels enclosed by the LV cavity mask in all the short-axis slices at the ED and ES images, taking into account the voxel size specified in the NIfTI header file. The LV ejection fraction is calculated using the formula $LVEF (\%) = (EDV - ESV) / EDV \times 100$. LV myocardial mass is calculated using the LV myocardial mass calculation formula: myocardium mask volume $\times$ myocardial density

1.05 g/mL (Leiner et al., 2019). The output of Agent 2 consists of the computed parameters (LVEF, EDV, ESV, LV mass) including their units, the dataset ID, and the flag indicating the level of the LVEF reduction (<40%=reduced, 40-50%=mildly reduced, >50%=preserved) according to ESC heart failure criteria.

3.5 Agent 3: Clinical Reasoning Agent

3.5.1 Design and Motivation

The Agent relies on a Retrieval Augmented Generation process that extracts medical passages from a curated database of cardiac MRI clinical guidelines and leverages these to ground the LLM’s diagnostic inference. In contrast to standard generation processes by an LLM, which are vulnerable to hallucinations and inconsistencies with clinical practice, this process guarantees that all diagnoses made are supported by a specific passage from a clinical guideline (Lewis et al., 2020; Pambudi et al., 2025).

3.5.2 Guideline Corpus Construction Pipeline

The corpus of guidelines consists of: (1) ESC 2022 Guidelines on Cardiac Magnetic Resonance Imaging; (2) ACC/AHA 2022 Guideline for the Diagnosis and Management of Heart Failure; (3) AHA/ACC 2020 Guideline for the Evaluation and Diagnosis of Chest Pain (cardiac imaging section); and (4) HFSA 2023 Comprehensive Heart Failure Practice Guideline. The documents are collected in their publicly available PDF format and then split into chunks containing 200-400 tokens each by the LangChain's RecursiveCharacterTextSplitter tool, with an overlap of 50 tokens between chunks to ensure context preservation. Next, the chunks are represented in the form of vectors with dimensions 768 using BioBERT model (Lee et al., 2020).

At inference time, Agent 3 takes the structured quantitative output from Agent 2 and builds a retrieval query representing the patient’s heart parameter information along with clinical interpretations (e.g., “LVEF 32%, reduced ejection fraction, DCM type”). FAISS index is queried for the k (k=5) closest semantically relevant guideline snippets using maximum inner product search. These guideline snippets are fed into a structured template along with the quantitative information, requesting the LLM to: (1) interpret the results based on the guideline snippets provided; (2) provide structured interpretation with clear citations from the guidelines; and (3) mark out any abnormal values according to the guidelines provided. The default choice of primary LLM used in this work is GPT-4 accessed through the OpenAI API; however, in the ablation experiment, another open-source LLM named MedLLaMA-3 is considered.

3.6 Agent 4: Explainability Agent

Two roles are fulfilled by the Explainability Agent: one being visualization via heatmaps based on the attention mechanism that is applied to the segmentation output and another role is providing explanation in the form of guidelines citations in the report.

Heatmaps are created using the Gradient-weighted Class Activation Mapping (Grad-CAM; Selvaraju et al., 2017) method on the feature maps from the segmentation network and then placed on the appropriate CMR short-axis images (ED frame). Grad-CAM method was chosen as a means to create heatmaps instead of SHAP or LIME because it allows for a direct

application to feature maps without any additional computations or model training steps (Haupt et al., 2025).

Regarding explainability for textual output, the cited references for each output by Agent 3 are arranged into citation-enhanced reports by Agent 4. Specifically, each diagnostic conclusion is linked to the original guidelines’ citations, facilitating verification of the rationale behind the AI-generated conclusions from primary guideline sources. The above explainability approach combines explainability via visuals for imaging findings and explainability for text for clinical decision-making. This solution tackles the gap identified in Chapter 1 regarding guideline-based explainability in the current generation of cardiac AI algorithms.

3.7 Datasets

There are three sets of data that are used for training, validation and generalisation testing. Their features are illustrated in Table 3.1. It should be emphasised that all data sets are public and do not need an ethics committee clearance for further analysis since they were issued by the corresponding challenge organisers using open-access licences.

Dataset	Year	Subjects	Classes	Modality	Purpose in This Study
ACDC	2018	150	5 pathologies	Cine CMR	Primary training & validation; segmentation + EF benchmarking
M&Ms	2021	375	4 pathologies	Cine CMR	Generalisation testing across 4 vendors and 6 clinical centres
M&Ms-2	2022	360	Multi-label	Cine CMR	Extended multi-label generalisation and stress-test evaluation

Table 3.1. Datasets used for training, validation, and generalisation testing.

3.8 Baseline Comparisons

An accurate comparative study necessitates the use of appropriately selected baselines, ranging from simple systems with single tasks to complex systems with multiple agents. The following table (Table 3.4) gives the definition of five systems used as baselines, with an appropriate explanation for their selections.

Baseline System	Type	Comparison Dimension	Rationale
nnU-Net Standalone	Single-task DL	Segmentation (DSC, HD95)	Strong segmentation baseline
CineMA	Foundation Model	Segmentation + EF (DSC, MAE)	Recent Cine CMR foundation model

MARL-Rad	Multi-agent Reporting	Report Quality (BERTScore, ROUGE)	Multi-agent report generation benchmark
CardAIc-Agents	Cardiac Multi-agent	End-to-end Performance (CAS, Accuracy)	Closest cardiac-specific agentic system

Table 3.4. Baseline systems for comparative evaluation, with comparison dimension and selection rationale.

3.11 Ethical Considerations

The experiments are strictly carried out using only public domain datasets (ACDC and M&Ms) which have been made openly accessible through research licenses. No recruitment of patients or collecting clinical data or even hospital data would be required in this project. Therefore, ethics clearance from the UMMC Ethics Committee would not be necessary for conducting experiments in this research.

Chapter 4: Expected outcomes

This research is expected to produce a clinically interpretable and generalisable agentic artificial intelligence framework for cardiac magnetic resonance imaging (CMR) analysis. The proposed system integrates automated segmentation, quantitative cardiac function assessment, and explainable AI techniques within a multi-agent architecture to support clinically meaningful decision-making.

A primary expected outcome is the development of a robust segmentation and quantification pipeline capable of accurately extracting left ventricular and right ventricular structures from cine CMR images and computing clinically relevant parameters including end-diastolic volume (EDV), end-systolic volume (ESV), and left ventricular ejection fraction (LVEF). The system is expected to achieve performance comparable to state-of-the-art methods on publicly available benchmark datasets such as ACDC and M&Ms.

Another expected outcome is the development of an explainability layer that combines visual explanations through Grad-CAM with guideline citation tracing. This dual explainability mechanism is expected to improve clinician trust by providing both image-level and knowledge-level justification for AI-generated findings.

From a research perspective, the study is expected to contribute a novel agentic architecture that integrates computer vision, quantitative cardiac assessment, retrieval-augmented reasoning, and explainable AI within a unified workflow. The framework will provide insights into how specialised AI agents can collaborate to perform complex medical image interpretation tasks.

Practically, the proposed system has the potential to reduce reporting workload, improve reporting consistency, and provide decision-support capabilities for clinicians in cardiac imaging workflows. The findings may contribute towards the development of trustworthy AI-assisted diagnostic systems for cardiovascular healthcare and support the broader adoption of explainable medical AI technologies.

Chapter 5: Preliminary Findings & Research Output

As part of the preliminary research undertaken in support of this proposal, a full experimental paper was developed and submitted to a peer-reviewed journal prior to the formal defence of this proposal. The paper, titled *Transparent Cardiac Intelligence: A Post-Hoc Explainability Framework for Ensemble-Based Arrhythmia Classification*, was submitted to **BMJ Medical Informatics and Decision Making** on 23 May 2026 and is currently under review.

5.1 Paper Description

The current work tackles the major deficiency in AI-based heart disease prediction, which is the lack of clarity in the diagnosis provided by such automatic systems. In order to bridge this gap, we design and assess an ensemble classification approach, which consists of multiple machine learning models to ensure more accurate arrhythmia detection using electrocardiograms. Notably, the system integrates the explainability approach after the prediction is made, using the approaches of SHAP and LIME (SHapley Additive exPlanations and Local Interpretable Model-agnostic Explanations).

5.2 Research Contribution

In relation to the overall hypothesis, this paper contributes to a number of areas. Firstly, it empirically confirms the potential of using ensemble learning techniques to provide competitive results in classifying cardiac data. Secondly, it confirms that XAI techniques can be applied after machine learning classification models for cardiac data, without affecting the predictive power of such models. Thirdly, it lays out the theoretical groundwork for the Explainability Agent introduced in Chapter 3.

The submission of this paper to a Q1-indexed medical informatics journal is indicative of the maturity of the preliminary research conducted and an indication of the academic value of the thesis idea.

Chapter 6: Work Schedule

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